Supplementary material

S1. Validation before the main experiments

CLAHE indicates contrast-limited adaptive histogram equalization; CLIP indicates Contrastive Language–Image Pretraining. CT indicates computed tomography.

S2. Data collection

Perception models generated findings reports before interpretation-model calls and were instructed to write at most 120 words. Median report lengths were 75 words for GPT-5 mini and Claude Haiku 4.5 and 90 for Gemini 3.7 Flash. Maximum lengths were 118, 121 and 127 words, respectively. Provider routing and token budgets remained fixed. Calls that failed with transient service errors were repeated until every item had a response. The automatic stop at more than 15% unparsed answers was not applied to blank-image conditions.

S3. Detailed results

Detailed results are reported with 95% confidence intervals (CIs).

S4. Open-weight models

S5. Text-only clinical questions

In a companion analysis (unpublished data), 8 language models answered text-only clinical questions from MedQA, MedAgentsBench and the text set of MedXpertQA. The models answered as single readers and in panels with specialty roles. The 340 configurations comprised 96 Independent, 96 Centralized, 96 Decentralized and 52 Hybrid panels, with architecture names from Kim et al.[1] Across the 288 Independent, Centralized and Decentralized configurations with 3 to 9 readers, mean initial-answer error correlation

Table S1: Validation checks and the settings they fixed.

Check	Result	Setting fixed
V1	Single-reader answer parsing was 100% for all four proprietary models called through OpenRouter. GPT-5 mini used 256 reasoning tokens at low effort and 2,304 at high effort on the same prompt.	GPT-5 mini at low effort.
V2	Single-reader accuracy relative to the original image, over 10 model-by-dataset cells: CLAHE 0.0 percentage points on average (range −6 to 14), grayscale inversion −6.6 (−12 to 2), 256 pixels −1.0, 128 pixels −5.6.	Inversion exceeded the limit of 5 percentage points. Contrast adjustment used gamma 0.7 for readers 4 to 6 and CLAHE for readers 7 to 9.
V3	With four images per prompt (the query and three reference cases), parsing was 100%. A CLIP cosine threshold of 0.95 removed about 32 similar cases from different patients per query. Radiographs from different patients reached cosines of 0.91 to 0.95. A perceptual hash alone missed the same CT level and the same patient's chest radiograph (cosine 0.99).	Reference-case exclusion rule as in Methods.
V4	In 20 inspected SLAKE-target cases, the target box covered the queried organ or lesion; the control box did not overlap it and lay within the body.	Used as built.
V5	Findings reports had a median of 72 to 77 words (maximum 109); answer parsing 100%.	Used as built.
V6	With option-letter probabilities, the LLaVA family put 96 to 100% of the probability on "A" regardless of the image. Pooled accuracy on 100 items (50 each from VQA-RAD and SLAKE): MedGemma 84%, HuatuoGPT-Vision 79%, Qwen3-VL 78%, Qwen2-VL 74%, LLaVA-1.5 55%, LLaVA-Med 53%.	Yes/no items read from the "yes"/"no" tokens; LLaVA-Med excluded by the 55% rule.
Throughput	Qwen 3.8 Flash used 250 to 6,400 reasoning tokens and 6 to 77 s per call.	Qwen 3.8 Flash was evaluated with original images, reduced resolution and blank images, self-consistency panels and contrast adjustments.

Table S2: Reading panels of 9 readers with the original images. Accuracy with item bootstrap 95% CI; pp, percentage points; initial, the answers before discussion.

Set	Model	n	Single (%)	Mean panel reader (%)	Independent panel (%)
VQA-RAD	GPT-5 mini	250	72.0 [66.4, 77.2]	71.3 [66.3, 76.2]	69.2 [63.2, 74.8]
VQA-RAD	Gemini 3.7 Flash	250	81.2 [76.4, 86.0]	81.3 [76.9, 85.4]	82.4 [77.2, 86.8]
VQA-RAD	Claude Haiku 4.5	250	60.8 [54.4, 66.8]	65.9 [61.0, 71.0]	64.8 [58.8, 70.8]
VQA-RAD	Qwen 3.8 Flash	250	69.6 [64.0, 75.2]	79.9 [76.3, 83.3]	83.2 [78.4, 87.6]
SLAKE	GPT-5 mini	250	80.8 [76.0, 85.6]	79.8 [75.4, 84.3]	78.4 [73.2, 83.6]
SLAKE	Gemini 3.7 Flash	250	91.2 [87.6, 94.4]	89.9 [86.7, 93.1]	90.0 [86.4, 93.6]
SLAKE	Claude Haiku 4.5	250	72.4 [67.2, 78.0]	72.3 [67.4, 77.0]	72.4 [66.8, 78.0]
SLAKE	Qwen 3.8 Flash	250	76.4 [70.8, 81.2]	83.3 [79.9, 87.0]	84.4 [80.0, 89.2]
MedXpertQA-MM	GPT-5 mini	250	54.8 [48.8, 60.4]	53.2 [48.1, 58.4]	55.6 [49.6, 61.6]
MedXpertQA-MM	Gemini 3.7 Flash	250	84.4 [79.6, 88.8]	83.3 [79.2, 87.2]	83.6 [78.8, 88.4]
MedXpertQA-MM	Claude Haiku 4.5	250	33.2 [27.6, 38.8]	30.8 [25.4, 36.1]	30.4 [24.4, 36.0]
MedXpertQA-MM	Qwen 3.8 Flash	250	43.6 [37.6, 49.6]	46.7 [41.9, 51.8]	49.2 [43.2, 55.2]

Table S3: Reading panels of 9 readers with the original images (continued).

Set	Model	Discussion panel (%)	Independent panel − single (pp)	Mean panel reader − single (pp)	Independent panel − its mean reader (pp)
VQA-RAD	GPT-5 mini	70.0 [64.0, 75.6]	−2.8 [−6.4, +0.8]	−0.7 [−3.5, +2.2]	−2.1 [−4.0, −0.4]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.0, 86.0]	+1.2 [−2.4, +4.8]	+0.1 [−2.7, +2.7]	+1.1 [−0.5, +2.7]
VQA-RAD	Claude Haiku 4.5	66.4 [60.4, 72.4]	+4.0 [−0.8, +8.8]	+5.1 [+1.0, +9.1]	−1.1 [−3.1, +0.8]
VQA-RAD	Qwen 3.8 Flash	80.4 [75.6, 85.2]	+13.6 [+7.6, +19.6]	+10.3 [+5.2, +15.5]	+3.3 [+0.8, +5.8]
SLAKE	GPT-5 mini	80.8 [75.6, 85.6]	−2.4 [−5.2, +0.4]	−1.0 [−3.5, +1.3]	−1.4 [−2.9, +0.0]
SLAKE	Gemini 3.7 Flash	89.2 [85.2, 93.2]	−1.2 [−3.2, +0.4]	−1.3 [−2.8, +0.1]	+0.1 [−1.0, +1.3]
SLAKE	Claude Haiku 4.5	74.4 [69.2, 79.6]	+0.0 [−4.4, +4.4]	−0.1 [−3.8, +3.6]	+0.1 [−1.6, +1.7]
SLAKE	Qwen 3.8 Flash	90.4 [86.8, 94.0]	+8.0 [+4.0, +12.4]	+6.9 [+3.5, +10.6]	+1.1 [−0.8, +2.9]
MedXpertQA-MM	GPT-5 mini	58.4 [52.4, 64.4]	+0.8 [−3.6, +5.6]	−1.6 [−5.2, +2.4]	+2.4 [+0.1, +4.6]
MedXpertQA-MM	Gemini 3.7 Flash	82.4 [77.6, 87.2]	−0.8 [−4.0, +2.4]	−1.1 [−3.5, +1.2]	+0.3 [−1.4, +2.0]
MedXpertQA-MM	Claude Haiku 4.5	30.0 [24.4, 36.0]	−2.8 [−6.0, +0.4]	−2.4 [−5.4, +0.5]	−0.4 [−1.9, +1.1]
MedXpertQA-MM	Qwen 3.8 Flash	52.4 [45.6, 58.8]	+5.6 [+0.4, +11.2]	+3.1 [−1.5, +7.6]	+2.5 [+0.0, +4.9]

Table S4: Reading panels of 9 readers with the original images (continued).

Set	Model	Discussion panel − single (pp)	ϕ initial	N_{eff}	Unanimous initial (%)
VQA-RAD	GPT-5 mini	−2.0 [−5.6, +1.6]	0.752 [0.697, 0.801]	1.28	74.4
VQA-RAD	Gemini 3.7 Flash	+0.0 [−3.6, +3.6]	0.725 [0.653, 0.785]	1.32	76.0
VQA-RAD	Claude Haiku 4.5	+5.6 [+0.8, +10.8]	0.731 [0.678, 0.781]	1.31	68.0
VQA-RAD	Qwen 3.8 Flash	+10.8 [+5.2, +16.4]	0.439 [0.369, 0.501]	1.99	54.8
SLAKE	GPT-5 mini	+0.0 [−2.8, +2.8]	0.785 [0.729, 0.844]	1.24	82.0
SLAKE	Gemini 3.7 Flash	−2.0 [−4.4, +0.0]	0.785 [0.696, 0.859]	1.24	90.4
SLAKE	Claude Haiku 4.5	+2.0 [−2.0, +6.0]	0.775 [0.723, 0.821]	1.25	75.6
SLAKE	Qwen 3.8 Flash	+14.0 [+9.2, +18.8]	0.529 [0.448, 0.597]	1.72	65.6
MedXpertQA-MM	GPT-5 mini	+3.6 [−0.8, +8.0]	0.685 [0.636, 0.735]	1.39	52.4
MedXpertQA-MM	Gemini 3.7 Flash	−2.0 [−5.2, +0.8]	0.749 [0.669, 0.817]	1.29	82.4
MedXpertQA-MM	Claude Haiku 4.5	−3.2 [−6.8, +0.4]	0.836 [0.790, 0.882]	1.17	71.6
MedXpertQA-MM	Qwen 3.8 Flash	+8.8 [+3.6, +14.4]	0.647 [0.591, 0.701]	1.46	41.2

Table S5: Reading panels of 9 readers with the original images (continued).

Set	Model	Wrong if unanimous initial (%)	ϕ final	Unanimous final (%)	Wrong if unanimous final (%)
VQA-RAD	GPT-5 mini	20.4	1.000 [1.000, 1.000]	100.0	30.0
VQA-RAD	Gemini 3.7 Flash	11.6	0.997 [0.991, 1.000]	99.6	18.9
VQA-RAD	Claude Haiku 4.5	25.9	0.996 [0.990, 1.000]	99.2	33.5
VQA-RAD	Qwen 3.8 Flash	5.8	0.822 [0.762, 0.874]	86.0	15.8
SLAKE	GPT-5 mini	12.2	0.995 [0.984, 1.000]	99.6	18.9
SLAKE	Gemini 3.7 Flash	5.8	1.000 [1.000, 1.000]	100.0	10.8
SLAKE	Claude Haiku 4.5	22.2	1.000 [1.000, 1.000]	100.0	25.6
SLAKE	Qwen 3.8 Flash	4.3	0.884 [0.811, 0.940]	94.4	8.9
MedXpertQA-MM	GPT-5 mini	35.9	0.996 [0.991, 1.000]	98.8	41.7
MedXpertQA-MM	Gemini 3.7 Flash	9.2	1.000 [1.000, 1.000]	100.0	17.6
MedXpertQA-MM	Claude Haiku 4.5	68.7	1.000 [1.000, 1.000]	100.0	70.0
MedXpertQA-MM	Qwen 3.8 Flash	36.9	0.913 [0.883, 0.941]	82.4	42.7

Table S6: Reduced resolution and blank images: single-reader accuracy (%) and error correlation ϕ of the independent panel of 9 in each image condition; paired differences between the original and the blank image.

Set	Model	Single $\leq$1024	Single 256	Single 128	Single blank
VQA-RAD	GPT-5 mini	72.0 [66.4, 77.2]	74.0 [68.8, 79.2]	72.8 [67.2, 78.0]	60.8 [54.8, 66.4]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.4, 86.0]	83.6 [78.8, 88.0]	77.6 [72.4, 82.4]	54.4 [48.4, 60.4]
VQA-RAD	Claude Haiku 4.5	60.8 [54.4, 66.8]	63.2 [57.2, 69.2]	65.2 [59.2, 70.8]	46.8 [40.8, 52.8]
VQA-RAD	Qwen 3.8 Flash	69.6 [64.0, 75.2]	67.6 [61.6, 73.2]	72.0 [66.4, 77.6]	54.8 [48.4, 60.8]
SLAKE	GPT-5 mini	80.8 [76.0, 85.6]	78.8 [73.6, 83.6]	75.2 [70.0, 80.4]	54.4 [48.4, 60.8]
SLAKE	Gemini 3.7 Flash	91.2 [87.6, 94.4]	91.2 [87.6, 94.8]	85.2 [80.8, 89.2]	51.2 [44.8, 57.6]
SLAKE	Claude Haiku 4.5	72.4 [67.2, 78.0]	73.6 [68.0, 78.8]	70.0 [64.4, 75.6]	38.8 [32.8, 44.8]
SLAKE	Qwen 3.8 Flash	76.4 [70.8, 81.2]	74.8 [69.2, 80.8]	79.2 [74.4, 84.0]	51.2 [44.8, 57.2]
MedXpertQA-MM	GPT-5 mini	54.8 [48.8, 60.4]	50.4 [44.4, 56.8]	50.8 [44.8, 56.8]	50.8 [44.8, 56.4]
MedXpertQA-MM	Gemini 3.7 Flash	84.4 [79.6, 88.8]	80.4 [75.2, 85.2]	78.4 [73.2, 83.2]	66.4 [60.8, 71.6]
MedXpertQA-MM	Claude Haiku 4.5	33.2 [27.6, 38.8]	32.0 [26.4, 37.6]	32.0 [26.4, 37.6]	29.6 [24.0, 35.2]
MedXpertQA-MM	Qwen 3.8 Flash	43.6 [37.6, 49.6]	40.8 [34.8, 47.2]	43.2 [37.2, 49.6]	41.6 [35.6, 47.6]

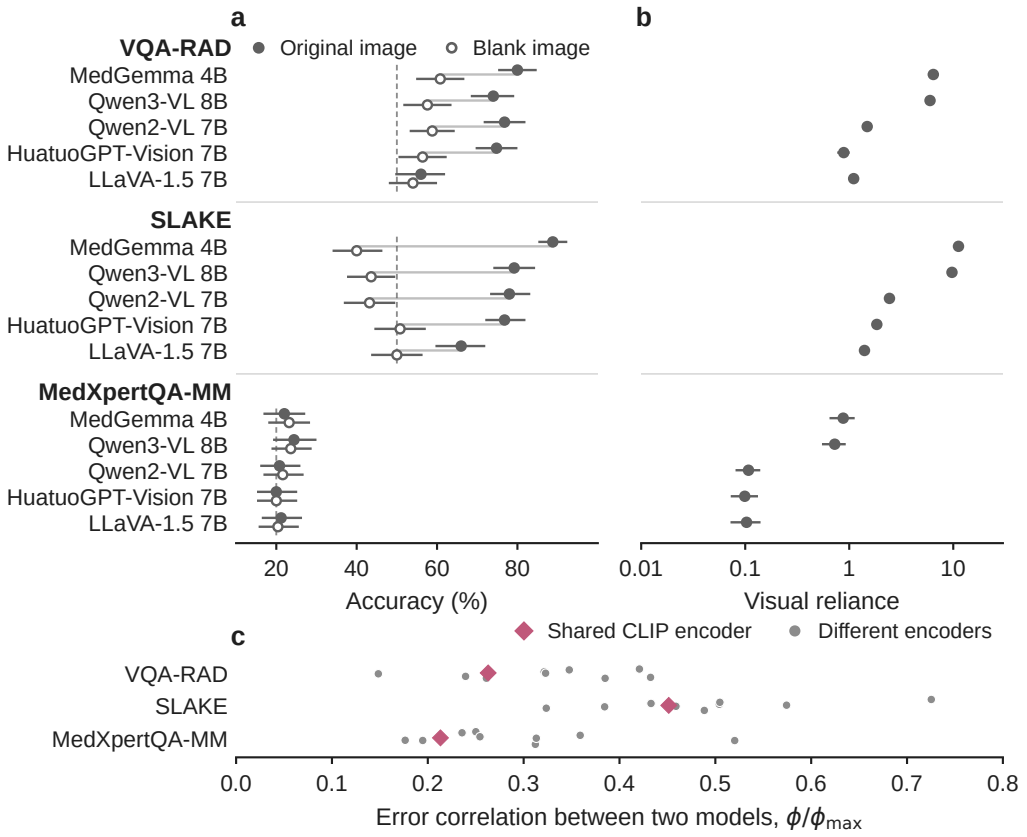


Figure S1: Open-weight models, one forward pass per item. (a) Accuracy labeled Original image (filled) and Blank image (open, a uniform gray image of the same size); dashed lines mark chance. (b) Visual reliance: absolute change in the log-odds of "yes" between the original and blank images on yes/no items. On MedXpertQA-MM, visual reliance was the Kullback–Leibler divergence from the original-image to blank-image probability distribution. Means with item bootstrap 95% CIs; log scale. (c) Error correlation between two models divided by its maximum for their error rates ($\phi/\phi_{\max}$). Diamonds mark HuatuoGPT-Vision 7B and LLaVA-1.5 7B, which share the CLIP vision encoder ViT-L/14-336.

Table S7: Reduced resolution and blank images (continued).

Set	Model	$\phi \leq 1024$	ϕ 256	ϕ 128	ϕ blank
VQA-RAD	GPT-5 mini	0.752 [0.697, 0.801]	0.694 [0.636, 0.751]	0.644 [0.586, 0.703]	0.742 [0.689, 0.789]
VQA-RAD	Gemini 3.7 Flash	0.725 [0.653, 0.785]	0.726 [0.646, 0.793]	0.749 [0.698, 0.804]	0.922 [0.889, 0.951]
VQA-RAD	Claude Haiku 4.5	0.731 [0.678, 0.781]	0.706 [0.651, 0.756]	0.591 [0.539, 0.650]	0.876 [0.840, 0.911]
VQA-RAD	Qwen 3.8 Flash	0.439 [0.369, 0.501]	0.442 [0.377, 0.501]	0.480 [0.415, 0.541]	0.841 [0.805, 0.879]
SLAKE	GPT-5 mini	0.785 [0.729, 0.844]	0.792 [0.741, 0.845]	0.709 [0.645, 0.761]	0.802 [0.755, 0.845]
SLAKE	Gemini 3.7 Flash	0.785 [0.696, 0.859]	0.803 [0.716, 0.869]	0.820 [0.753, 0.872]	0.982 [0.968, 0.994]
SLAKE	Claude Haiku 4.5	0.775 [0.723, 0.821]	0.752 [0.695, 0.811]	0.718 [0.664, 0.768]	0.879 [0.841, 0.915]
SLAKE	Qwen 3.8 Flash	0.529 [0.448, 0.597]	0.547 [0.479, 0.607]	0.490 [0.408, 0.560]	0.831 [0.788, 0.866]
MedXpertQA-MM	GPT-5 mini	0.685 [0.636, 0.735]	0.690 [0.637, 0.739]	0.700 [0.652, 0.752]	0.730 [0.679, 0.780]
MedXpertQA-MM	Gemini 3.7 Flash	0.749 [0.669, 0.817]	0.762 [0.698, 0.823]	0.784 [0.715, 0.841]	0.799 [0.748, 0.844]
MedXpertQA-MM	Claude Haiku 4.5	0.836 [0.790, 0.882]	0.821 [0.776, 0.868]	0.840 [0.795, 0.881]	0.818 [0.765, 0.861]
MedXpertQA-MM	Qwen 3.8 Flash	0.647 [0.591, 0.701]	0.630 [0.573, 0.688]	0.609 [0.556, 0.667]	0.656 [0.602, 0.709]

was 0.758. These architectures shared the initial answers and had identical initial-answer error correlations by design. With 9 readers, error correlation was 0.762, of the same magnitude as the imaging values. The effective number of independent readers (N_{eff}) was 1.27. Dispersion across configurations was described by the standard deviation (SD). Across the 96 Decentralized configurations, error correlation was 0.758 (SD 0.117) before discussion and 0.990 (SD 0.010) after discussion. After discussion, N_{eff} was 1.01.

References

[1] Y. Kim, K. Gu, C. Park, C. Park, S. Schmidgall, A. A. Heydari, Y. Yan, Z. Zhang, Y. Zhuang, Y. Liu, M. Malhotra, P. P. Liang, H. W. Park, Y. Yang, X. Xu, Y. Du, S. Patel, T. Althoff, D. McDuff, X. Liu, Capable language models can outgrow the benefits of collaboration, Nature Machine Intelligence 8 (2026) 1157–1172, preprint: arXiv:2512.08296, *Towards a Science of Scaling Agent Systems*. doi:10.1038/s42256-026-01268-y.

Table S8: Reduced resolution and blank images (continued).

Set	Model	Single original − blank (pp)	Independent panel original − blank (pp)	Abstentions, blank (%)	ϕ blank, answered items
VQA-RAD	GPT-5 mini	+11.2 [+4.0, +18.4]	+7.6 [+0.4, +14.8]	–	–
VQA-RAD	Gemini 3.7 Flash	+26.8 [+18.8, +35.2]	+28.4 [+20.8, +36.8]	–	–
VQA-RAD	Claude Haiku 4.5	+14.0 [+6.0, +22.0]	+10.4 [+2.8, +18.4]	10.4	0.963 [0.942, 0.982]
VQA-RAD	Qwen 3.8 Flash	+14.8 [+6.4, +23.2]	+26.0 [+18.4, +34.0]	0.1	–
SLAKE	GPT-5 mini	+26.4 [+18.8, +33.2]	+25.6 [+18.0, +33.2]	–	–
SLAKE	Gemini 3.7 Flash	+40.0 [+33.2, +46.8]	+39.2 [+32.4, +46.0]	–	–
SLAKE	Claude Haiku 4.5	+33.6 [+24.0, +42.8]	+28.4 [+19.6, +37.2]	4.3	0.931 [0.901, 0.956]
SLAKE	Qwen 3.8 Flash	+25.2 [+16.4, +33.6]	+33.6 [+25.6, +42.0]	–	–
MedXpertQA-MM	GPT-5 mini	+4.0 [−2.0, +10.0]	+6.8 [+1.6, +12.0]	–	–
MedXpertQA-MM	Gemini 3.7 Flash	+18.0 [+12.0, +23.6]	+17.6 [+12.0, +23.6]	–	–
MedXpertQA-MM	Claude Haiku 4.5	+3.6 [−0.4, +8.0]	+0.0 [−4.4, +4.4]	14.8	0.852 [0.803, 0.891]
MedXpertQA-MM	Qwen 3.8 Flash	+2.0 [−4.4, +8.0]	+9.6 [+4.8, +14.8]	0.5	–

Table S9: Reader answers of the independent panel of 9 cross-tabulated by correctness with the original and with the blank image (% of answers), and shared wrong answers (at least 7 of 9 readers wrong) whose majority answer matched the blank-image majority.

Set	Model	Right with original and blank	Right only with original	Wrong only with original	Wrong with original and blank
VQA-RAD	GPT-5 mini	47.2	24.1	12.0	16.7
VQA-RAD	Gemini 3.7 Flash	44.0	37.3	10.4	8.3
VQA-RAD	Claude Haiku 4.5	36.6	29.3	13.6	20.5
VQA-RAD	Qwen 3.8 Flash	46.0	33.9	10.8	9.3
SLAKE	GPT-5 mini	43.5	36.3	9.5	10.7
SLAKE	Gemini 3.7 Flash	46.7	43.2	4.4	5.8
SLAKE	Claude Haiku 4.5	25.9	46.4	15.8	11.9
SLAKE	Qwen 3.8 Flash	38.5	44.8	12.0	4.7
MedXpertQA-MM	GPT-5 mini	39.7	13.5	7.9	38.9
MedXpertQA-MM	Gemini 3.7 Flash	61.0	22.3	4.4	12.3
MedXpertQA-MM	Claude Haiku 4.5	22.7	8.1	5.2	64.0
MedXpertQA-MM	Qwen 3.8 Flash	32.9	13.8	7.0	46.4

Table S10: Reader answers of the independent panel of 9 cross-tabulated by correctness with the original and with the blank image (% of answers), and shared wrong answers (at least 7 of 9 readers wrong) whose majority answer matched the blank-image majority (continued).

Set	Model	Shared wrong answers (n)	Shared wrong answers matching the blank-image majority (%)
VQA-RAD	GPT-5 mini	57	54.4 [42.1, 66.7]
VQA-RAD	Gemini 3.7 Flash	35	42.9 [25.7, 60.0]
VQA-RAD	Claude Haiku 4.5	70	54.3 [42.9, 65.7]
VQA-RAD	Qwen 3.8 Flash	18	44.4 [22.2, 66.7]
SLAKE	GPT-5 mini	42	57.1 [42.9, 71.4]
SLAKE	Gemini 3.7 Flash	19	57.9 [36.8, 78.9]
SLAKE	Claude Haiku 4.5	56	33.9 [21.4, 46.4]
SLAKE	Qwen 3.8 Flash	21	14.3 [0.0, 28.6]
MedXpertQA-MM	GPT-5 mini	95	64.2 [54.7, 72.7]
MedXpertQA-MM	Gemini 3.7 Flash	31	74.2 [58.1, 90.3]
MedXpertQA-MM	Claude Haiku 4.5	163	71.2 [64.4, 77.9]
MedXpertQA-MM	Qwen 3.8 Flash	105	76.2 [67.6, 83.8]

Table S11: Region occlusion on SLAKE-target: share of answers changed by covering the target box or an equal-size control box, and occlusion sensitivity (target minus control).

Model	Models	Target answer change (%)	Control answer change (%)	Occlusion sensitivity (pp)	Single target answer change (%)
GPT-5 mini	Proprietary, panel readers	23.5 [19.7, 27.6]	6.4 [4.7, 8.4]	+17.1 [+13.2, +21.2]	26.4 [20.8, 32.0]
Gemini 3.7 Flash	Proprietary, panel readers	36.4 [31.6, 41.4]	3.6 [2.4, 4.9]	+32.9 [+28.2, +38.0]	36.8 [30.8, 42.8]
Claude Haiku 4.5	Proprietary, panel readers	10.7 [8.2, 13.6]	7.5 [5.5, 9.6]	+3.2 [+0.8, +5.7]	8.4 [5.2, 12.0]
HuatuoGPT-Vision 7B	Open-weight	28.0 [22.4, 34.0]	6.4 [3.6, 9.6]	+21.6 [+15.6, +28.0]	
LLaVA-1.5 7B	Open-weight	12.8 [8.8, 17.2]	10.4 [6.8, 14.4]	+2.4 [−2.0, +6.8]	
MedGemma 4B	Open-weight	13.2 [9.2, 17.6]	1.6 [0.4, 3.2]	+11.6 [+7.6, +15.6]	
Qwen2-VL 7B	Open-weight	11.2 [7.6, 15.2]	4.8 [2.4, 7.6]	+6.4 [+2.0, +10.4]	
Qwen3-VL 8B	Open-weight	23.6 [18.4, 28.8]	3.6 [1.6, 6.0]	+20.0 [+14.4, +26.0]	

Table S12: Region occlusion on SLAKE-target (continued).

Model	Models	Single control answer change (%)	Single occlusion sensitivity (pp)
GPT-5 mini	Proprietary, panel readers	8.4 [5.2, 12.0]	+18.0 [+11.6, +24.8]
Gemini 3.7 Flash	Proprietary, panel readers	2.8 [1.2, 4.8]	+34.0 [+28.0, +40.4]
Claude Haiku 4.5	Proprietary, panel readers	6.4 [3.6, 9.2]	+2.0 [−1.6, +5.2]
HuatuoGPT-Vision 7B	Open-weight		
LLaVA-1.5 7B	Open-weight		
MedGemma 4B	Open-weight		
Qwen2-VL 7B	Open-weight		
Qwen3-VL 8B	Open-weight		

Table S13: Paired images (SLAKE-pairs, 200 pairs of images with the same question and opposite reference answers).

Model	Single: both correct (%)	Single: answers differ (%)	Independent panel: both correct (%)	Independent panel: answers differ (%)
GPT-5 mini	59.0 [52.0, 65.5]	62.0 [55.5, 68.5]	62.0 [55.5, 68.5]	65.0 [58.0, 71.0]
Gemini 3.7 Flash	74.0 [67.5, 80.0]	75.0 [69.0, 81.0]	74.5 [68.0, 80.5]	75.5 [69.0, 81.5]
Claude Haiku 4.5	38.5 [31.5, 45.0]	42.5 [35.5, 49.5]	42.5 [35.5, 49.0]	46.0 [39.0, 52.5]
HuatuoGPT-Vision 7B (open weight)	48.5 [41.5, 55.5]	53.0 [46.0, 60.0]		
LLaVA-1.5 7B (open weight)	31.5 [25.5, 38.0]	39.5 [33.0, 46.5]		
MedGemma 4B (open weight)	62.5 [56.0, 68.5]	63.5 [57.0, 69.5]		
Qwen2-VL 7B (open weight)	49.5 [42.5, 56.0]	53.5 [46.5, 60.0]		
Qwen3-VL 8B (open weight)	53.5 [46.5, 60.5]	56.5 [49.5, 63.0]		

Table S14: Paired images (SLAKE-pairs, 200 pairs of images with the same question and opposite reference answers) (continued).

Model	Readers: answers differ (%)
GPT-5 mini	66.0 [60.4, 71.2]
Gemini 3.7 Flash	75.9 [70.1, 81.4]
Claude Haiku 4.5	45.3 [39.2, 51.3]
HuatuoGPT-Vision 7B (open weight)	
LLaVA-1.5 7B (open weight)	
MedGemma 4B (open weight)	
Qwen2-VL 7B (open weight)	
Qwen3-VL 8B (open weight)	

Table S15: Discussion panels: conformity, the share of readers correct in their initial answer who gave a wrong final answer, by whether the initial answer differed from the same reader's blank-image answer and by whether it was in the initial majority.

Set	Model	Conformity, differs from blank (%)	Conformity, same as blank (%)	Difference (pp)	Conformity, initial minority (%)
VQA-RAD	GPT-5 mini	9.8 [5.5, 14.9]	4.6 [2.3, 7.4]	+5.1 [+0.4, +10.4]	72.3 [55.2, 88.9]
VQA-RAD	Gemini 3.7 Flash	3.7 [1.4, 6.2]	3.4 [1.6, 5.7]	+0.3 [−2.9, +3.3]	100.0 [100.0, 100.0]
VQA-RAD	Claude Haiku 4.5	7.1 [3.9, 11.0]	5.7 [3.1, 8.9]	+1.4 [−2.9, +5.9]	75.0 [59.2, 89.7]
VQA-RAD	Qwen 3.8 Flash	12.5 [7.8, 17.7]	6.6 [3.8, 9.8]	+5.9 [+0.6, +11.5]	54.1 [37.4, 70.6]
SLAKE	GPT-5 mini	4.5 [2.1, 7.3]	1.9 [0.8, 3.3]	+2.6 [+0.3, +5.2]	56.1 [34.9, 78.8]
SLAKE	Gemini 3.7 Flash	2.9 [0.7, 5.3]	1.4 [0.3, 2.9]	+1.5 [−1.1, +4.2]	71.4 [41.2, 100.0]
SLAKE	Claude Haiku 4.5	4.0 [1.9, 6.7]	3.4 [1.1, 6.4]	+0.6 [−3.3, +4.1]	62.5 [41.4, 83.0]
SLAKE	Qwen 3.8 Flash	4.2 [2.0, 6.9]	2.8 [1.3, 4.6]	+1.4 [−1.4, +4.6]	41.5 [23.7, 61.2]
MedXpertQA-MM	GPT-5 mini	18.1 [11.5, 26.1]	2.7 [1.4, 4.3]	+15.4 [+9.6, +23.1]	68.5 [52.6, 85.3]
MedXpertQA-MM	Gemini 3.7 Flash	7.2 [3.7, 11.6]	2.6 [1.2, 4.5]	+4.6 [+0.8, +9.0]	71.7 [48.0, 93.9]
MedXpertQA-MM	Claude Haiku 4.5	23.6 [12.8, 36.3]	6.3 [2.9, 11.1]	+17.4 [+6.4, +29.7]	83.9 [67.3, 100.0]
MedXpertQA-MM	Qwen 3.8 Flash	18.1 [12.3, 25.1]	5.4 [2.9, 8.5]	+12.7 [+6.6, +19.8]	62.2 [48.6, 76.7]

Table S16: Discussion panels (continued).

Set	Model	Conformity, initial majority (%)	Differs from blank: in initial minority (%)	Same as blank: in initial minority (%)	n correct (differs/same)
VQA-RAD	GPT-5 mini	1.9 [0.4, 3.8]	11.2 [6.8, 16.4]	3.8 [2.0, 5.9]	543/1062
VQA-RAD	Gemini 3.7 Flash	1.0 [0.0, 2.3]	2.5 [0.8, 4.5]	2.6 [1.1, 4.5]	840/989
VQA-RAD	Claude Haiku 4.5	1.4 [0.0, 3.2]	8.5 [5.2, 12.8]	5.3 [3.0, 8.0]	660/823
VQA-RAD	Qwen 3.8 Flash	6.5 [3.9, 9.4]	6.2 [3.4, 9.4]	4.9 [2.6, 7.7]	762/1035
SLAKE	GPT-5 mini	1.1 [0.0, 2.5]	4.0 [2.1, 6.4]	3.4 [1.7, 5.5]	817/978
SLAKE	Gemini 3.7 Flash	1.2 [0.2, 2.4]	1.8 [0.5, 3.5]	1.0 [0.3, 2.1]	971/1051
SLAKE	Claude Haiku 4.5	1.4 [0.3, 2.9]	3.4 [1.8, 5.7]	4.8 [2.0, 8.3]	1044/583
SLAKE	Qwen 3.8 Flash	1.8 [0.7, 3.1]	3.4 [1.6, 5.5]	5.5 [3.3, 8.5]	1009/866
MedXpertQA-MM	GPT-5 mini	1.4 [0.0, 3.4]	20.1 [13.3, 28.4]	3.5 [1.8, 5.5]	304/894
MedXpertQA-MM	Gemini 3.7 Flash	1.9 [0.6, 3.5]	6.6 [3.3, 11.0]	1.5 [0.6, 2.6]	501/1373
MedXpertQA-MM	Claude Haiku 4.5	4.4 [0.9, 9.1]	19.8 [11.1, 31.0]	3.9 [1.7, 6.9]	182/511
MedXpertQA-MM	Qwen 3.8 Flash	2.4 [0.7, 4.5]	23.5 [16.3, 32.1]	6.2 [3.6, 9.4]	310/740

Table S17: Perception and interpretation: mean reader accuracy (%) when each of 9 readers reads its own query-agnostic findings report from a perception model, against reading the image; error correlation ϕ.

Set	Interpretation model	Reports by GPT-5 mini	Reports by Gemini	Reports by Haiku	Image
VQA-RAD	GPT-5 mini	69.0 [63.7, 73.9]	78.7 [74.1, 82.8]	61.6 [56.4, 66.7]	72.6 [67.6, 77.3]
VQA-RAD	Gemini 3.7 Flash	70.2 [64.6, 75.2]	81.3 [76.9, 85.4]	63.6 [58.4, 68.6]	80.3 [75.6, 84.5]
VQA-RAD	Claude Haiku 4.5	70.6 [65.4, 75.6]	78.8 [74.2, 83.1]	63.5 [58.1, 68.4]	62.5 [56.7, 68.0]
SLAKE	GPT-5 mini	83.5 [79.3, 87.5]	87.4 [83.7, 90.9]	78.8 [74.3, 82.9]	79.1 [74.5, 83.6]
SLAKE	Gemini 3.7 Flash	84.9 [81.0, 88.7]	88.2 [84.6, 91.6]	80.2 [76.0, 84.4]	89.7 [86.3, 93.1]
SLAKE	Claude Haiku 4.5	81.0 [76.8, 85.1]	84.3 [80.4, 88.1]	76.3 [71.8, 80.8]	73.4 [68.1, 78.5]
MedXpertQA-MM	GPT-5 mini	50.1 [45.0, 55.2]	62.0 [56.7, 67.2]	45.0 [40.1, 49.9]	52.9 [47.7, 58.0]
MedXpertQA-MM	Gemini 3.7 Flash	60.8 [55.6, 65.9]	74.5 [69.6, 79.1]	57.8 [52.7, 62.7]	83.7 [79.4, 87.8]
MedXpertQA-MM	Claude Haiku 4.5	30.6 [25.6, 35.7]	45.6 [40.0, 50.9]	27.4 [22.6, 32.2]	31.7 [26.3, 37.3]

Table S18: Perception and interpretation (continued).

Set	Interpretation model	GPT-5 mini reports − image (pp)	Gemini reports − image (pp)	Haiku reports − image (pp)	ϕ image
VQA-RAD	GPT-5 mini	−3.6 [−8.4, +1.4]	+6.1 [+1.4, +10.9]	−10.9 [−16.8, −5.0]	0.741 [0.683, 0.797]
VQA-RAD	Gemini 3.7 Flash	−10.1 [−16.1, −4.0]	+1.0 [−4.0, +6.0]	−16.7 [−23.4, −10.1]	0.776 [0.715, 0.836]
VQA-RAD	Claude Haiku 4.5	+8.1 [+1.4, +14.9]	+16.3 [+9.9, +22.6]	+0.9 [−5.3, +6.8]	0.878 [0.835, 0.911]
SLAKE	GPT-5 mini	+4.4 [+0.9, +8.2]	+8.3 [+4.4, +12.4]	−0.4 [−5.0, +4.6]	0.800 [0.746, 0.844]
SLAKE	Gemini 3.7 Flash	−4.8 [−8.5, −1.3]	−1.6 [−4.0, +1.2]	−9.5 [−13.9, −4.9]	0.828 [0.744, 0.895]
SLAKE	Claude Haiku 4.5	+7.6 [+3.1, +12.1]	+10.9 [+6.4, +15.7]	+2.9 [−1.6, +7.3]	0.882 [0.838, 0.922]
MedXpertQA-MM	GPT-5 mini	−2.8 [−6.9, +1.0]	+9.1 [+3.4, +14.4]	−7.9 [−12.7, −3.2]	0.692 [0.642, 0.742]
MedXpertQA-MM	Gemini 3.7 Flash	−22.9 [−28.4, −17.5]	−9.2 [−13.3, −5.2]	−25.9 [−31.2, −20.4]	0.811 [0.742, 0.866]
MedXpertQA-MM	Claude Haiku 4.5	−1.1 [−5.9, +3.6]	+13.8 [+7.8, +19.5]	−4.4 [−9.0, −0.0]	0.915 [0.876, 0.949]

Table S19: Perception and interpretation (continued).

Set	Interpretation model	ϕ own-report condition (same-model reports)	ϕ shared-report condition
VQA-RAD	GPT-5 mini	0.818 [0.770, 0.866]	0.944 [0.917, 0.970]
VQA-RAD	Gemini 3.7 Flash	0.783 [0.727, 0.841]	0.990 [0.975, 1.000]
VQA-RAD	Claude Haiku 4.5	0.723 [0.668, 0.773]	0.986 [0.973, 0.998]
SLAKE	GPT-5 mini	0.777 [0.711, 0.842]	0.955 [0.921, 0.985]
SLAKE	Gemini 3.7 Flash	0.771 [0.687, 0.840]	0.965 [0.934, 0.988]
SLAKE	Claude Haiku 4.5	0.728 [0.665, 0.785]	0.969 [0.943, 0.990]
MedXpertQA-MM	GPT-5 mini	0.641 [0.595, 0.696]	0.800 [0.755, 0.845]
MedXpertQA-MM	Gemini 3.7 Flash	0.715 [0.654, 0.770]	0.880 [0.831, 0.927]
MedXpertQA-MM	Claude Haiku 4.5	0.701 [0.644, 0.758]	0.893 [0.857, 0.933]

Table S20: Error correlation by shared component, VQA-RAD: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); self-consistency, the error correlation ϕ of self-consistency readers of one model.

Component	Model or model pair	$\Delta\phi$ (self-consistency: ϕ)
Self-consistency (ϕ)	Claude Haiku 4.5	0.878 [0.836, 0.918]
Self-consistency (ϕ)	Gemini 3.7 Flash	0.776 [0.714, 0.825]
Self-consistency (ϕ)	GPT-5 mini	0.741 [0.676, 0.792]
Self-consistency (ϕ)	Qwen 3.8 Flash	0.442 [0.378, 0.504]
Role	Mean over models	−0.048 [−0.071, −0.023]
Role	Claude Haiku 4.5	−0.148 [−0.198, −0.098]
Role	Gemini 3.7 Flash	−0.051 [−0.104, −0.002]
Role	GPT-5 mini	+0.011 [−0.029, +0.053]
Role	Qwen 3.8 Flash	−0.003 [−0.049, +0.042]
Contrast adjustment	Mean over models	−0.092 [−0.123, −0.068]
Contrast adjustment	Claude Haiku 4.5	−0.151 [−0.211, −0.101]
Contrast adjustment	Gemini 3.7 Flash	−0.181 [−0.260, −0.125]
Contrast adjustment	GPT-5 mini	−0.061 [−0.103, −0.014]
Contrast adjustment	Qwen 3.8 Flash	+0.023 [−0.003, +0.048]
Reference cases	Mean over models	−0.110 [−0.142, −0.084]
Reference cases	Claude Haiku 4.5	−0.229 [−0.273, −0.191]
Reference cases	Gemini 3.7 Flash	−0.043 [−0.093, +0.001]
Reference cases	GPT-5 mini	−0.058 [−0.102, −0.023]
Report	Mean over models	−0.199 [−0.233, −0.164]
Report	Claude Haiku 4.5	−0.263 [−0.314, −0.212]
Report	Gemini 3.7 Flash	−0.207 [−0.266, −0.151]
Report	GPT-5 mini	−0.126 [−0.170, −0.085]
Perception model	Mean over models	−0.311 [−0.362, −0.260]
Perception model	Claude Haiku 4.5	−0.315 [−0.369, −0.256]
Perception model	Gemini 3.7 Flash	−0.322 [−0.375, −0.264]
Perception model	GPT-5 mini	−0.298 [−0.354, −0.248]
Interpretation model	Mean over models	−0.108 [−0.132, −0.083]
Interpretation model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.132 [−0.167, −0.104]
Interpretation model	Claude Haiku 4.5 and GPT-5 mini	−0.125 [−0.159, −0.093]
Interpretation model	Gemini 3.7 Flash and GPT-5 mini	−0.066 [−0.098, −0.039]
Model	Mean over models	−0.470 [−0.522, −0.421]

Table S21: Error correlation by shared component, SLAKE: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); self-consistency, the error correlation ϕ of self-consistency readers of one model.

Component	Model or model pair	$\Delta\phi$ (self-consistency: ϕ)
Self-consistency (ϕ)	Claude Haiku 4.5	0.882 [0.836, 0.921]
Self-consistency (ϕ)	Gemini 3.7 Flash	0.828 [0.754, 0.906]
Self-consistency (ϕ)	GPT-5 mini	0.800 [0.751, 0.847]
Self-consistency (ϕ)	Qwen 3.8 Flash	0.604 [0.530, 0.666]
Role	Mean over models	−0.060 [−0.087, −0.036]
Role	Claude Haiku 4.5	−0.107 [−0.149, −0.069]
Role	Gemini 3.7 Flash	−0.043 [−0.106, +0.007]
Role	GPT-5 mini	−0.015 [−0.051, +0.022]
Role	Qwen 3.8 Flash	−0.075 [−0.139, −0.016]
Contrast adjustment	Mean over models	−0.094 [−0.123, −0.066]
Contrast adjustment	Claude Haiku 4.5	−0.101 [−0.150, −0.054]
Contrast adjustment	Gemini 3.7 Flash	−0.108 [−0.204, −0.043]
Contrast adjustment	GPT-5 mini	−0.138 [−0.199, −0.090]
Contrast adjustment	Qwen 3.8 Flash	−0.028 [−0.065, +0.011]
Reference cases	Mean over models	−0.146 [−0.179, −0.117]
Reference cases	Claude Haiku 4.5	−0.247 [−0.292, −0.204]
Reference cases	Gemini 3.7 Flash	−0.111 [−0.174, −0.054]
Reference cases	GPT-5 mini	−0.082 [−0.125, −0.038]
Report	Mean over models	−0.204 [−0.247, −0.162]
Report	Claude Haiku 4.5	−0.242 [−0.308, −0.181]
Report	Gemini 3.7 Flash	−0.193 [−0.274, −0.125]
Report	GPT-5 mini	−0.179 [−0.256, −0.115]
Perception model	Mean over models	−0.275 [−0.334, −0.218]
Perception model	Claude Haiku 4.5	−0.236 [−0.296, −0.179]
Perception model	Gemini 3.7 Flash	−0.303 [−0.373, −0.244]
Perception model	GPT-5 mini	−0.287 [−0.352, −0.225]
Interpretation model	Mean over models	−0.167 [−0.222, −0.129]
Interpretation model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.224 [−0.297, −0.166]
Interpretation model	Claude Haiku 4.5 and GPT-5 mini	−0.194 [−0.265, −0.144]
Interpretation model	Gemini 3.7 Flash and GPT-5 mini	−0.081 [−0.134, −0.043]
Model	Mean over models	−0.473 [−0.544, −0.406]

Table S22: Error correlation by shared component, MedXpertQA-MM: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); self-consistency, the error correlation ϕ of self-consistency readers of one model.

Component	Model or model pair	$\Delta\phi$ (self-consistency: ϕ)
Self-consistency (ϕ)	Claude Haiku 4.5	0.915 [0.885, 0.949]
Self-consistency (ϕ)	Gemini 3.7 Flash	0.811 [0.743, 0.863]
Self-consistency (ϕ)	GPT-5 mini	0.692 [0.642, 0.739]
Self-consistency (ϕ)	Qwen 3.8 Flash	0.629 [0.570, 0.685]
Role	Mean over models	−0.033 [−0.051, −0.014]
Role	Claude Haiku 4.5	−0.080 [−0.119, −0.041]
Role	Gemini 3.7 Flash	−0.062 [−0.114, −0.018]
Role	GPT-5 mini	−0.007 [−0.039, +0.022]
Role	Qwen 3.8 Flash	+0.017 [−0.019, +0.054]
Contrast adjustment	Mean over models	−0.045 [−0.062, −0.027]
Contrast adjustment	Claude Haiku 4.5	−0.089 [−0.136, −0.048]
Contrast adjustment	Gemini 3.7 Flash	−0.080 [−0.131, −0.033]
Contrast adjustment	GPT-5 mini	−0.015 [−0.041, +0.009]
Contrast adjustment	Qwen 3.8 Flash	+0.005 [−0.016, +0.028]
Report	Mean over models	−0.172 [−0.211, −0.142]
Report	Claude Haiku 4.5	−0.192 [−0.250, −0.144]
Report	Gemini 3.7 Flash	−0.165 [−0.218, −0.110]
Report	GPT-5 mini	−0.159 [−0.213, −0.104]
Perception model	Mean over models	−0.263 [−0.294, −0.227]
Perception model	Claude Haiku 4.5	−0.278 [−0.324, −0.227]
Perception model	Gemini 3.7 Flash	−0.249 [−0.292, −0.203]
Perception model	GPT-5 mini	−0.263 [−0.308, −0.218]
Interpretation model	Mean over models	−0.430 [−0.477, −0.389]
Interpretation model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.546 [−0.605, −0.488]
Interpretation model	Claude Haiku 4.5 and GPT-5 mini	−0.435 [−0.500, −0.379]
Interpretation model	Gemini 3.7 Flash and GPT-5 mini	−0.310 [−0.371, −0.256]
Model	Mean over models	−0.492 [−0.536, −0.445]
Model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.689 [−0.790, −0.614]
Model	Claude Haiku 4.5 and GPT-5 mini	−0.483 [−0.567, −0.394]
Model	Claude Haiku 4.5 and Qwen 3.8 Flash	−0.472 [−0.560, −0.398]
Model	Gemini 3.7 Flash and GPT-5 mini	−0.493 [−0.580, −0.417]

Table S23: Regression of pairwise error correlation on the components two readers do not share (additive; one intercept per pair of interpretation models), a secondary summary of the matched contrasts.

Set	Perception model	Image vs report	Contrast adjustment	Role
VQA-RAD	−0.247 [−0.283, −0.210]	−0.123 [−0.173, −0.078]	−0.023 [−0.036, −0.009]	−0.006 [−0.016, +0.005]
SLAKE	−0.193 [−0.232, −0.146]	−0.094 [−0.135, −0.060]	−0.017 [−0.032, −0.003]	−0.004 [−0.015, +0.006]
MedXpertQA-MM	−0.160 [−0.187, −0.131]	−0.099 [−0.127, −0.071]	−0.012 [−0.021, −0.003]	−0.009 [−0.017, −0.002]

Table S24: Regression of pairwise error correlation on the components two readers do not share (additive; one intercept per pair of interpretation models), a secondary summary of the matched contrasts (continued).

Set	Reference cases	Report	Interpretation model (mean)
VQA-RAD	−0.026 [−0.043, −0.011]	+0.150 [+0.092, +0.218]	−0.026 [−0.045, −0.004]
SLAKE	−0.040 [−0.053, −0.028]	+0.009 [−0.051, +0.059]	−0.124 [−0.168, −0.086]
MedXpertQA-MM	+0.000 [+0.000, +0.000]	−0.066 [−0.104, −0.022]	−0.194 [−0.228, −0.164]

Table S25: Reference cases (retrieved in-context examples): accuracy (%) and error correlation ϕ with no reference cases, shared reference cases, and private reference cases, with images or as text.

Set	Model	Single, none (%)	Single, shared images (%)	Single, shared text (%)	Panel, none (%)
VQA-RAD	GPT-5 mini	72.0 [66.4, 77.2]	73.6 [68.0, 78.8]	75.6 [70.0, 80.4]	73.6 [68.0, 78.8]
SLAKE	GPT-5 mini	80.8 [76.0, 85.6]	79.2 [74.4, 84.0]	81.6 [76.8, 86.4]	78.4 [73.2, 83.6]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.4, 86.0]	82.4 [77.2, 87.2]	83.6 [78.8, 88.0]	80.4 [75.2, 84.8]
SLAKE	Gemini 3.7 Flash	91.2 [87.6, 94.4]	89.2 [85.2, 92.8]	90.8 [87.2, 94.4]	90.0 [86.4, 93.6]
VQA-RAD	Claude Haiku 4.5	60.8 [54.4, 66.8]	68.8 [63.2, 74.0]	61.2 [54.8, 67.2]	62.0 [55.6, 68.0]
SLAKE	Claude Haiku 4.5	72.4 [67.2, 78.0]	67.6 [62.0, 73.2]	76.8 [71.6, 82.0]	73.6 [68.0, 79.2]

Table S26: Reference cases (retrieved in-context examples) (continued).

Set	Model	Panel, shared images (%)	Panel, private images (%)	Panel, shared text (%)	Panel, private text (%)
VQA-RAD	GPT-5 mini	74.8 [69.6, 80.4]	75.2 [69.6, 80.4]	76.4 [71.2, 81.6]	74.8 [69.6, 80.0]
SLAKE	GPT-5 mini	81.6 [76.8, 86.4]	84.0 [79.2, 88.4]	82.0 [77.2, 86.8]	81.6 [76.8, 86.4]
VQA-RAD	Gemini 3.7 Flash	82.8 [78.4, 87.2]	82.0 [77.2, 86.4]	84.4 [80.0, 88.8]	84.4 [79.6, 88.8]
SLAKE	Gemini 3.7 Flash	89.2 [85.6, 92.8]	90.4 [86.8, 93.6]	90.4 [86.4, 94.0]	92.0 [88.8, 95.2]
VQA-RAD	Claude Haiku 4.5	65.2 [58.8, 71.2]	64.4 [58.8, 70.0]	62.4 [56.0, 68.4]	62.4 [56.0, 68.0]
SLAKE	Claude Haiku 4.5	70.4 [64.8, 76.0]	68.8 [62.8, 74.8]	78.0 [73.2, 83.2]	78.0 [73.2, 82.8]

Table S27: Reference cases (retrieved in-context examples) (continued).

Set	Model	ϕ none	ϕ shared images	ϕ private images	ϕ shared text
VQA-RAD	GPT-5 mini	0.741 [0.683, 0.797]	0.751 [0.694, 0.799]	0.668 [0.608, 0.723]	0.737 [0.681, 0.792]
SLAKE	GPT-5 mini	0.800 [0.746, 0.844]	0.741 [0.667, 0.807]	0.667 [0.600, 0.737]	0.780 [0.718, 0.842]
VQA-RAD	Gemini 3.7 Flash	0.776 [0.715, 0.836]	0.708 [0.640, 0.772]	0.673 [0.599, 0.744]	0.743 [0.658, 0.816]
SLAKE	Gemini 3.7 Flash	0.828 [0.744, 0.895]	0.836 [0.761, 0.894]	0.722 [0.629, 0.797]	0.823 [0.725, 0.891]
VQA-RAD	Claude Haiku 4.5	0.878 [0.835, 0.911]	0.890 [0.849, 0.928]	0.664 [0.612, 0.713]	0.912 [0.878, 0.942]
SLAKE	Claude Haiku 4.5	0.882 [0.838, 0.922]	0.928 [0.894, 0.955]	0.655 [0.608, 0.708]	0.941 [0.907, 0.970]

Table S28: Reference cases (retrieved in-context examples) (continued).

Set	Model	ϕ private text
VQA-RAD	GPT-5 mini	0.703 [0.637, 0.759]
SLAKE	GPT-5 mini	0.690 [0.620, 0.752]
VQA-RAD	Gemini 3.7 Flash	0.692 [0.618, 0.763]
SLAKE	Gemini 3.7 Flash	0.716 [0.599, 0.804]
VQA-RAD	Claude Haiku 4.5	0.681 [0.630, 0.739]
SLAKE	Claude Haiku 4.5	0.720 [0.657, 0.779]

Table S29: Open-weight models: accuracy (%) and visual reliance (|change in log-odds of yes| between the original and the blank image on yes/no items; KL divergence on MedXpertQA-MM).

Model	Set	Accuracy, original (%)	Accuracy, blank (%)	256 px	128 px
HuatuoGPT-Vision 7B	VQA-RAD	74.8 [69.6, 80.0]	56.4 [50.4, 62.4]	77.2 [72.0, 82.4]	75.6 [70.4, 80.8]
HuatuoGPT-Vision 7B	SLAKE	76.8 [72.0, 82.0]	50.8 [44.4, 57.2]	76.0 [70.8, 81.2]	76.0 [70.8, 81.6]
HuatuoGPT-Vision 7B	MedXpertQA-MM	20.0 [15.2, 25.2]	20.0 [15.2, 25.2]	19.6 [15.2, 24.4]	20.4 [15.6, 25.2]
LLaVA-1.5 7B	VQA-RAD	56.0 [49.6, 62.0]	54.0 [48.0, 60.0]	56.4 [50.4, 62.4]	56.0 [49.6, 62.0]
LLaVA-1.5 7B	SLAKE	66.0 [59.6, 72.0]	50.0 [43.6, 56.4]	66.4 [60.4, 72.4]	66.0 [60.4, 72.0]
LLaVA-1.5 7B	MedXpertQA-MM	21.2 [16.4, 26.4]	20.4 [15.6, 25.6]	21.2 [16.4, 26.4]	20.8 [16.0, 25.6]
MedGemma 4B	VQA-RAD	80.0 [75.2, 84.8]	60.8 [54.8, 66.8]	80.4 [75.6, 85.2]	76.0 [71.2, 81.2]
MedGemma 4B	SLAKE	88.8 [85.2, 92.4]	40.0 [34.0, 46.4]	86.8 [82.8, 91.2]	84.0 [79.6, 88.4]
MedGemma 4B	MedXpertQA-MM	22.0 [16.8, 27.2]	23.2 [18.0, 28.4]	22.4 [17.2, 27.6]	22.4 [17.2, 27.6]
Qwen2-VL 7B	VQA-RAD	76.8 [71.6, 82.0]	58.8 [53.2, 64.4]	73.6 [68.4, 78.8]	76.4 [70.8, 81.6]
Qwen2-VL 7B	SLAKE	78.0 [73.2, 83.2]	43.2 [36.8, 49.6]	72.8 [67.6, 78.4]	68.0 [62.4, 74.0]
Qwen2-VL 7B	MedXpertQA-MM	20.8 [16.0, 26.0]	21.6 [16.8, 26.8]	21.2 [16.4, 26.4]	21.2 [16.4, 26.4]
Qwen3-VL 8B	VQA-RAD	74.0 [68.4, 79.2]	57.6 [51.6, 63.6]	70.0 [64.0, 75.6]	68.8 [62.8, 74.4]
Qwen3-VL 8B	SLAKE	79.2 [74.0, 84.4]	43.6 [37.6, 49.6]	83.6 [78.8, 88.0]	81.2 [76.4, 86.0]
Qwen3-VL 8B	MedXpertQA-MM	24.4 [19.2, 30.0]	23.6 [18.8, 28.8]	25.6 [20.4, 31.2]	25.2 [20.0, 30.4]

Table S30: Open-weight models (continued).

Model	Set	Visual reliance, mean	median
HuatuoGPT-Vision 7B	VQA-RAD	0.89 [0.77, 1.01]	0.62
HuatuoGPT-Vision 7B	SLAKE	1.84 [1.66, 2.03]	1.50
HuatuoGPT-Vision 7B	MedXpertQA-MM	0.10 [0.07, 0.13]	0.04
LLaVA-1.5 7B	VQA-RAD	1.10 [1.02, 1.18]	1.12
LLaVA-1.5 7B	SLAKE	1.40 [1.29, 1.52]	1.25
LLaVA-1.5 7B	MedXpertQA-MM	0.10 [0.07, 0.14]	0.02
MedGemma 4B	VQA-RAD	6.43 [5.71, 7.17]	4.38
MedGemma 4B	SLAKE	11.23 [10.58, 11.95]	11.63
MedGemma 4B	MedXpertQA-MM	0.88 [0.65, 1.13]	0.02
Qwen2-VL 7B	VQA-RAD	1.49 [1.33, 1.64]	1.12
Qwen2-VL 7B	SLAKE	2.44 [2.21, 2.66]	2.13
Qwen2-VL 7B	MedXpertQA-MM	0.11 [0.08, 0.14]	0.04
Qwen3-VL 8B	VQA-RAD	5.98 [5.36, 6.61]	4.38
Qwen3-VL 8B	SLAKE	9.72 [8.95, 10.55]	9.07
Qwen3-VL 8B	MedXpertQA-MM	0.72 [0.55, 0.93]	0.07

Table S31: Open-weight models: error correlation between two models.

Set	Model 1	Model 2	ϕ	$\phi/\phi_{\max}$	Shared encoder
VQA-RAD	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.172	0.263	yes
VQA-RAD	HuatuoGPT-Vision 7B	MedGemma 4B	0.332	0.385	no
VQA-RAD	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.227	0.239	no
VQA-RAD	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.412	0.421	no
VQA-RAD	LLaVA-1.5 7B	MedGemma 4B	0.181	0.321	no
VQA-RAD	LLaVA-1.5 7B	Qwen2-VL 7B	0.162	0.261	no
VQA-RAD	LLaVA-1.5 7B	Qwen3-VL 8B	0.099	0.148	no
VQA-RAD	MedGemma 4B	Qwen2-VL 7B	0.294	0.323	no
VQA-RAD	MedGemma 4B	Qwen3-VL 8B	0.365	0.432	no
VQA-RAD	Qwen2-VL 7B	Qwen3-VL 8B	0.322	0.348	no
SLAKE	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.346	0.451	yes
SLAKE	HuatuoGPT-Vision 7B	MedGemma 4B	0.316	0.488	no
SLAKE	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.371	0.385	no
SLAKE	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.535	0.574	no
SLAKE	LLaVA-1.5 7B	MedGemma 4B	0.227	0.459	no
SLAKE	LLaVA-1.5 7B	Qwen2-VL 7B	0.373	0.504	no
SLAKE	LLaVA-1.5 7B	Qwen3-VL 8B	0.360	0.505	no
SLAKE	MedGemma 4B	Qwen2-VL 7B	0.485	0.725	no
SLAKE	MedGemma 4B	Qwen3-VL 8B	0.224	0.324	no
SLAKE	Qwen2-VL 7B	Qwen3-VL 8B	0.418	0.433	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.205	0.213	yes
MedXpertQA-MM	HuatuoGPT-Vision 7B	MedGemma 4B	0.338	0.359	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.507	0.520	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.275	0.312	no
MedXpertQA-MM	LLaVA-1.5 7B	MedGemma 4B	0.244	0.250	no
MedXpertQA-MM	LLaVA-1.5 7B	Qwen2-VL 7B	0.192	0.195	no
MedXpertQA-MM	LLaVA-1.5 7B	Qwen3-VL 8B	0.161	0.176	no
MedXpertQA-MM	MedGemma 4B	Qwen2-VL 7B	0.227	0.236	no
MedXpertQA-MM	MedGemma 4B	Qwen3-VL 8B	0.238	0.254	no
MedXpertQA-MM	Qwen2-VL 7B	Qwen3-VL 8B	0.282	0.313	no